

Movie Analytics Data Warehouse

Multi-Source Reconciliation & Financial/Critical Performance Analysis

Students:

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A reproducible Data Warehouse integrating heterogeneous movie sources for multidimensional performance analysis.



Project Objectives & Business Requirements

The Data Warehouse supports multidimensional analysis of financial performance, audience reception and the evolution of cinema over time.

01

Financial Performance

Which genres, periods and movies generate the highest revenue and profit?

Budget → Revenue → Profit

02

Audience vs Critical Reception

How do IMDb and TMDB ratings differ across movies and budget ranges?

IMDb Rating ↔ TMDB Rating

03

Temporal Evolution

How do movie production and financial performance evolve across decades?

Day → Month → Year → Decade

Additional Analytical Requirements

The Data Warehouse is extended with three additional OLAP questions using the same multidimensional model.

04

Runtime & Profitability

Does movie duration relate to average profit and IMDb rating?

Runtime → Profit + IMDb Rating

05

Audience Engagement

Is audience attention associated with movie profitability?

IMDb Votes → Profit

06

Geographic Distribution

How do movie volume and average rating vary across continents?

Continent → Volume + IMDb Rating

Data Sources & Heterogeneity

The project combines heterogeneous movie datasets that differ in structure, identifiers, volume and data completeness.

SOURCE 01

Kaggle Dataset

CSV-based movie data containing identifiers and analytical attributes such as budget, revenue and ratings.

CSV • Structured

SOURCE 02

IMDb Non Commercial

IMDb datasets distributed as TSV files, with movie metadata, ratings and title information.

TSV • Structured

Main Data Challenges

Missing Values

Incomplete fields require cleaning and reconciliation before analysis.

Different Volumes

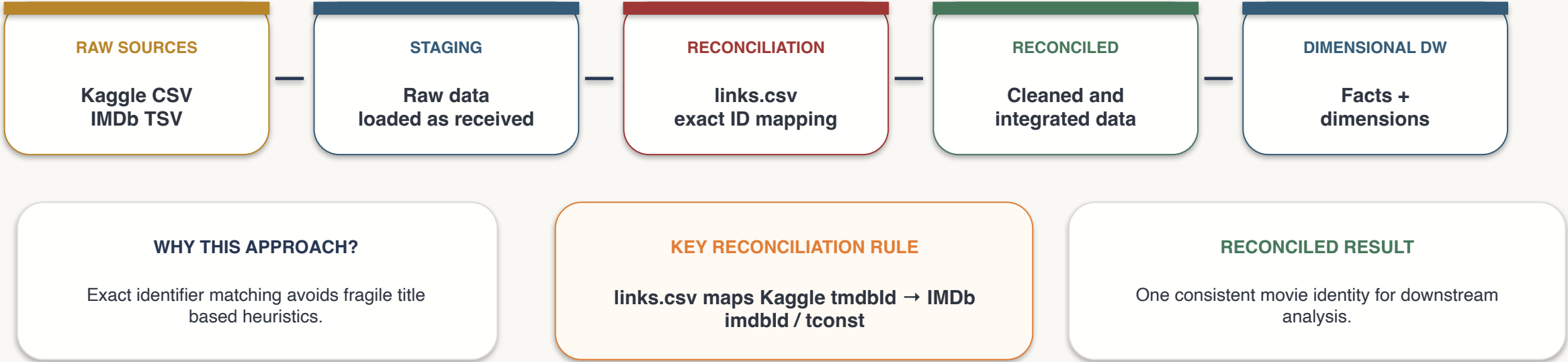
The sources contain different amounts of information and coverage.

Non Uniform Formats

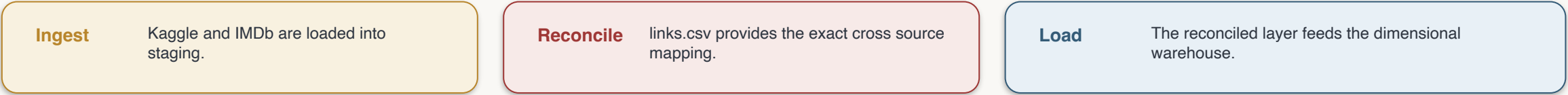
CSV and TSV structures, fields and representations are not directly aligned.

ETL Pipeline & Reconciled Layer

The pipeline transforms heterogeneous raw sources into a consistent analytical layer before loading the dimensional warehouse.

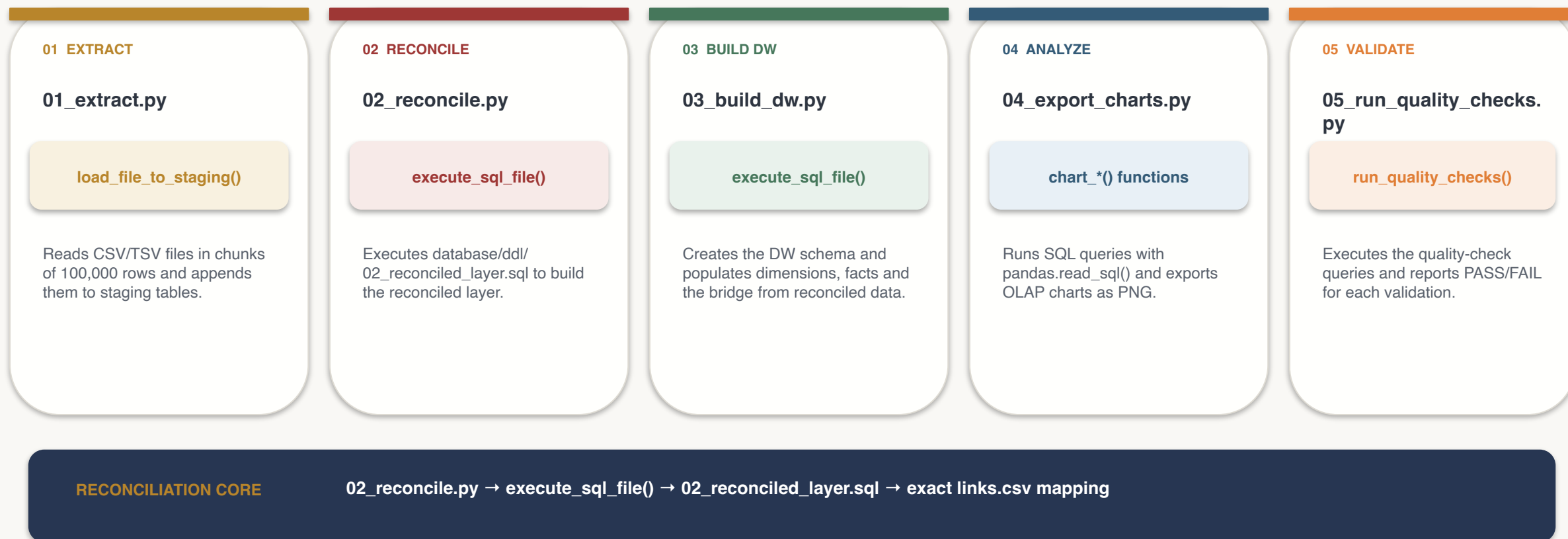


Pipeline Logic



ETL Implementation: Functions & Operations

Each Python script orchestrates one stage of the pipeline; the data transformations themselves are executed through PostgreSQL SQL scripts.



Note: the Python scripts orchestrate the pipeline; detailed joins, cleaning rules and identifier reconciliation are defined in the referenced SQL files.

Conceptual Design: Dimensional Fact Model

The DFM defines the analytical grain of the fact and the dimensions used to analyse movie performance.

Fact Granularity

1 record = 1 movie performance

The fact stores movie-level measures for financial and critical-performance analysis.

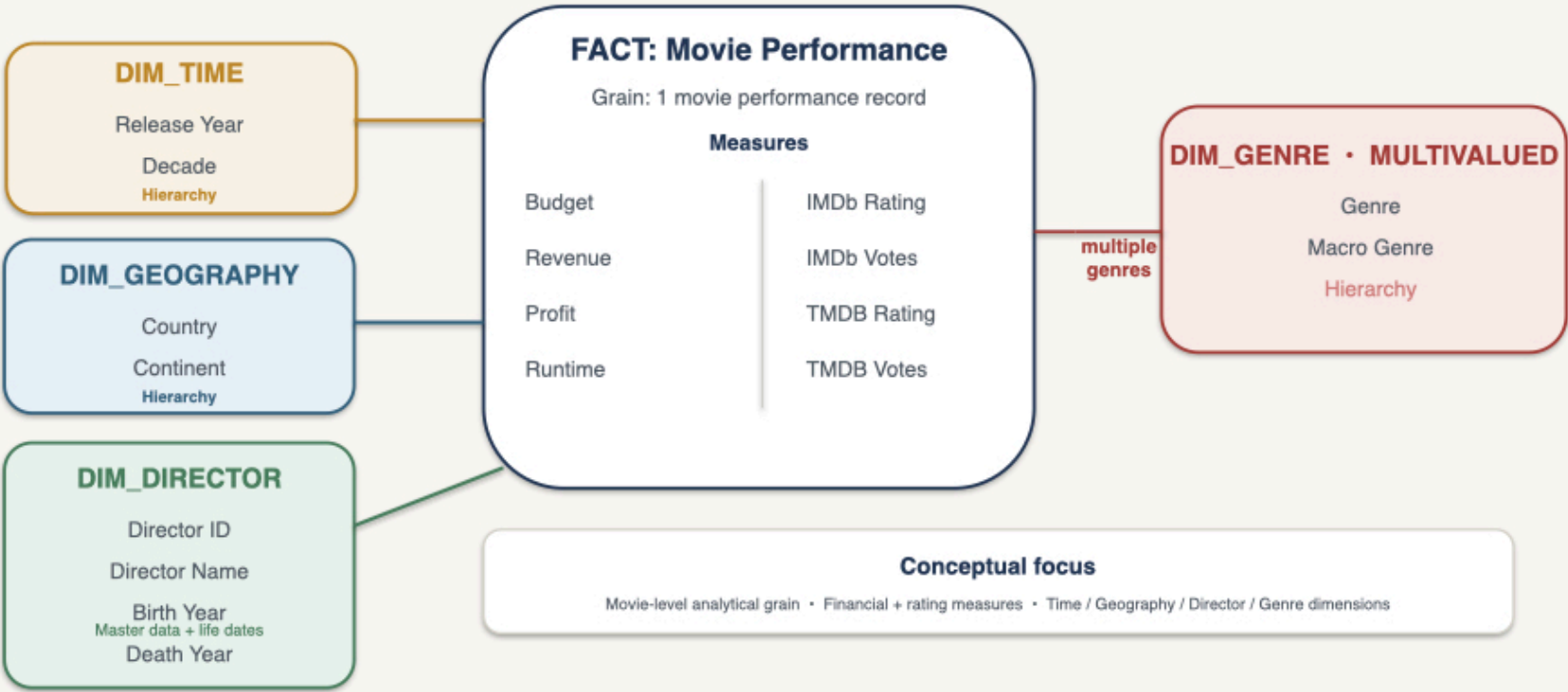
Main Dimensions

Time	Release Year → Decade
Geography	Country → Continent
Director	Master data + life dates
Genre	Genre → Macro Genre

CONCEPTUAL DESIGN • DIMENSIONAL FACT MODEL

Movie Analytics — Conceptual DFM

Movie-level fact grain with analytical dimensions and explicit hierarchies



Logical Snowflake Schema & Bridge Table

Normalized dimensions with an explicit bridge for the multivalued genre relationship

Relational Schema

FACT TABLE

fact_movie_performance is the central table.
Each row represents one movie performance and stores the analytical measures.

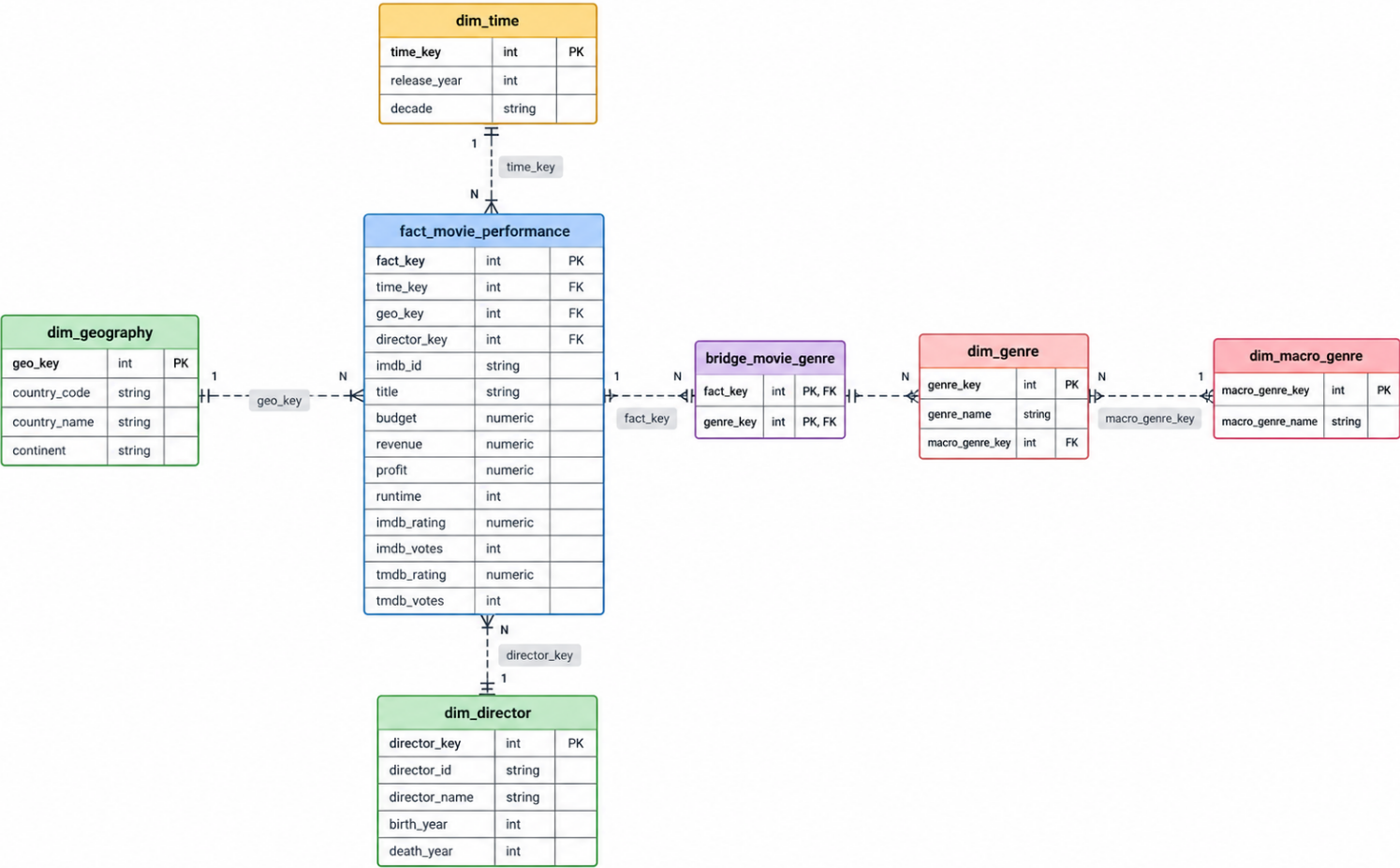
NORMALIZED DIMENSIONS

Time → release_year → decade
Geography → country → continent
Director → director master data
Genre → genre → macro genre

BRIDGE TABLE

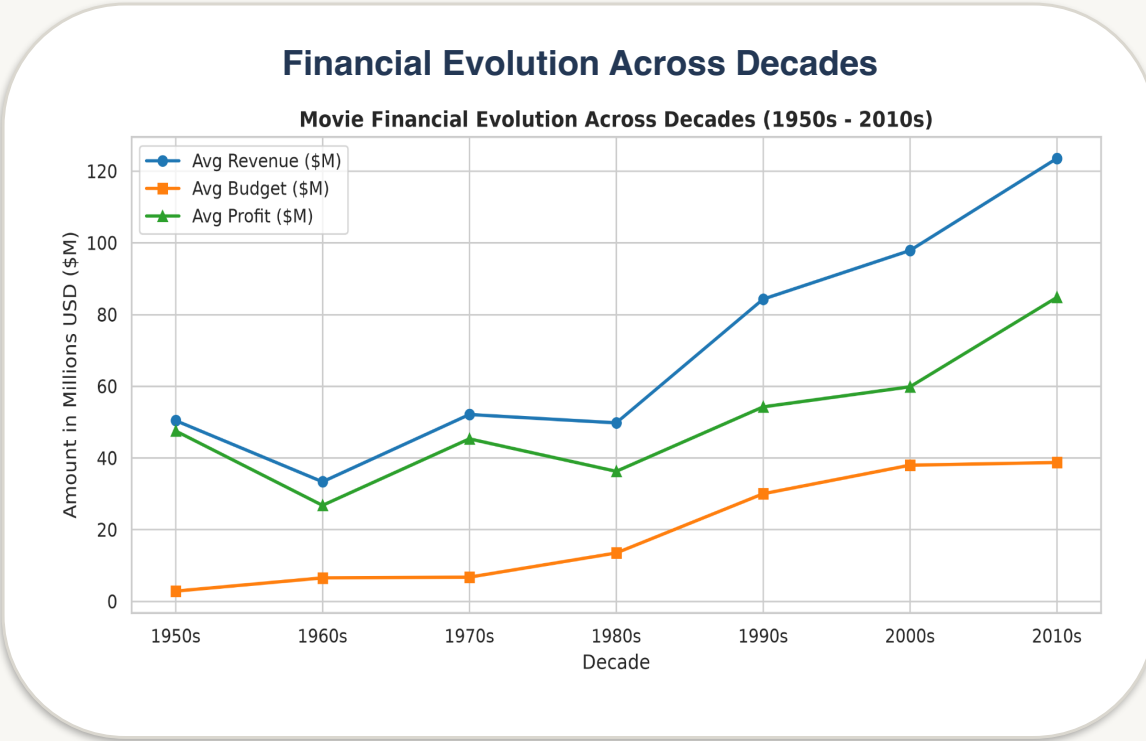
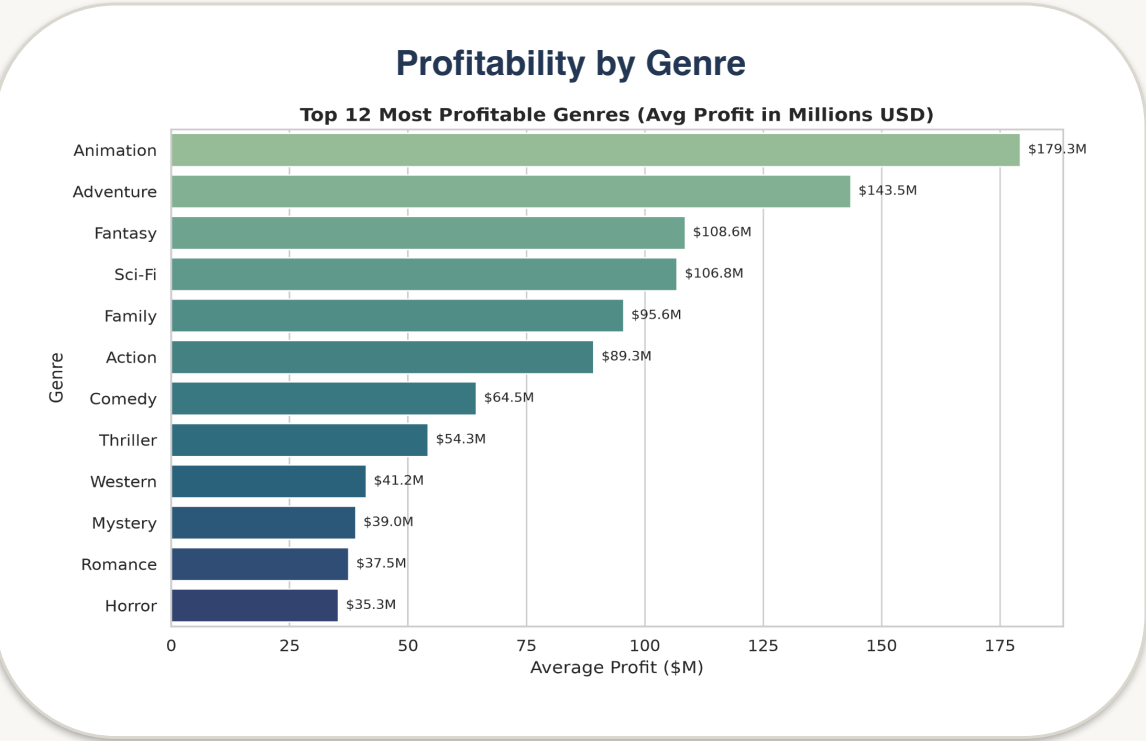
bridge_movie_genre resolves the multivalued relationship: one movie can belong to multiple genres. The key is (fact_key, genre_key).

Controlled normalization + analytical flexibility



OLAP Analysis: Financial & Temporal Trends

Profitability by genre and the evolution of financial performance across decades



QUERY 1 — GENRE

The aggregation identifies the genres with the highest average profit, after requiring at least 50 movies per genre. Animation has the highest average profit in this result.

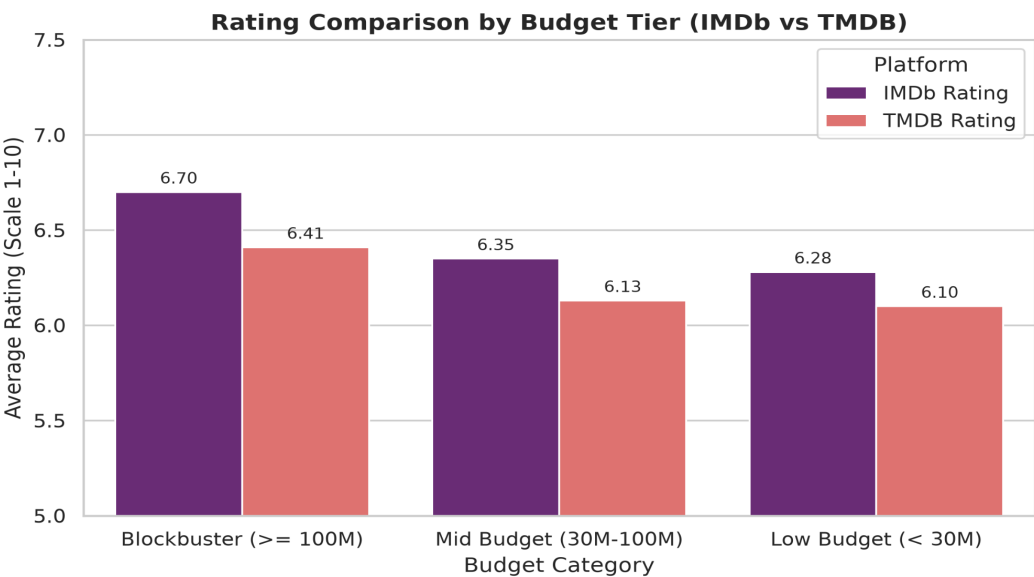
QUERY 2 — DECADES

Average revenue, budget and profit generally increase toward the most recent decades, with a visible acceleration from the 1980s to the 1990s and onward.

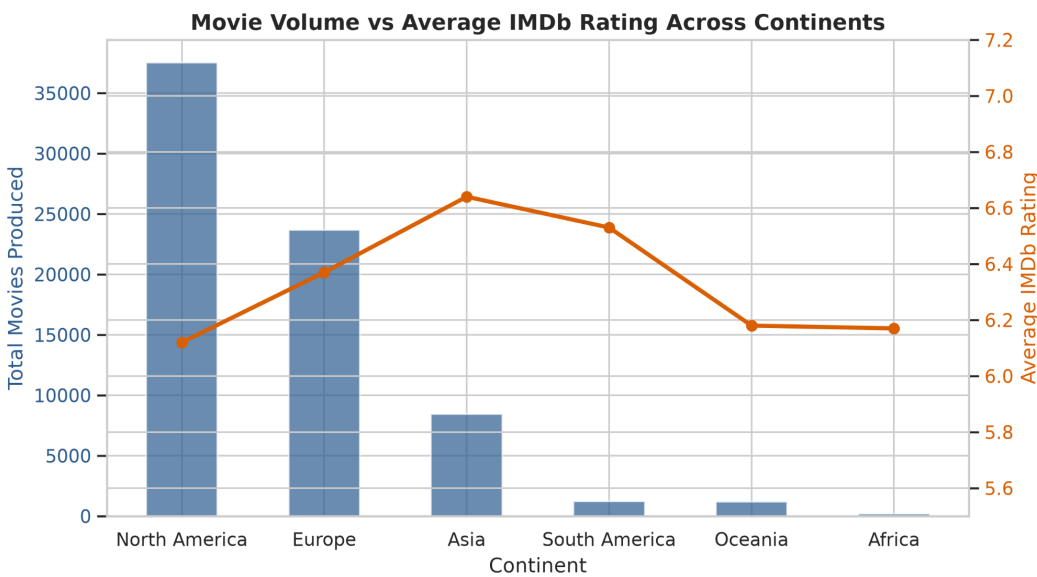
OLAP Analysis: Ratings & Geographic Distribution

Comparison of audience ratings across platforms and geographic production patterns

IMDb vs TMDB by Budget Tier



Movie Volume & IMDb Rating by Continent



QUERY 3 — RATING COMPARISON

IMDb ratings are consistently higher than TMDB ratings across all budget tiers. The difference is visible for both low- and high-budget movies.

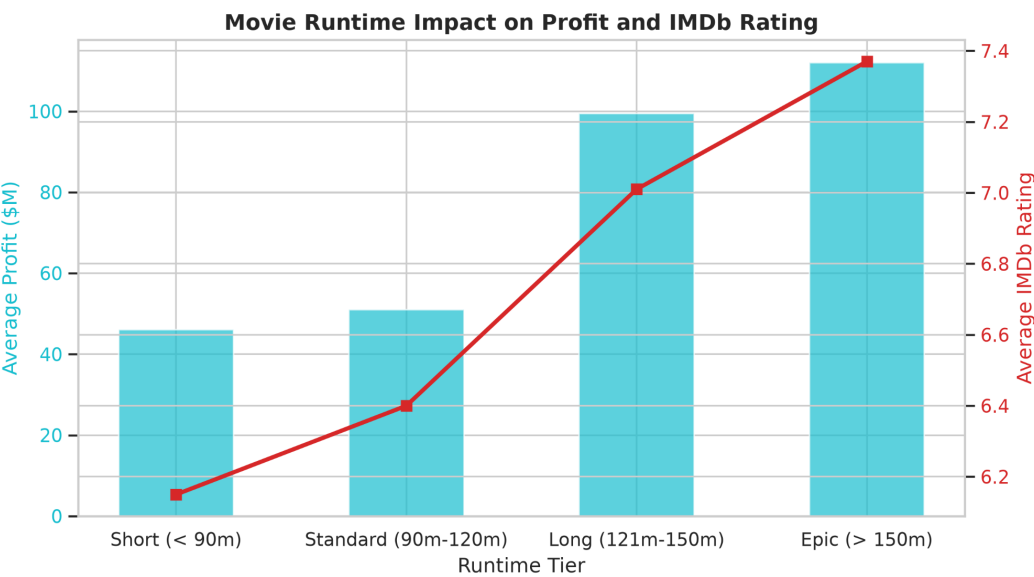
QUERY 4 — GEOGRAPHIC DISTRIBUTION

North America and Europe account for the largest movie volumes in the dataset, while average IMDb ratings vary across continents and do not follow production volume directly.

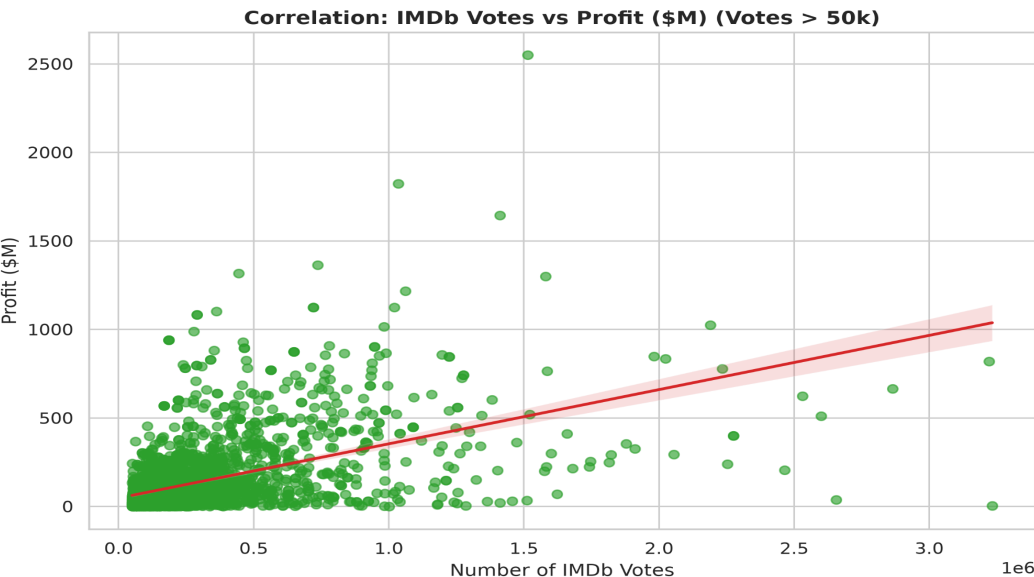
OLAP Analysis: Runtime, Engagement & Profitability

Two complementary queries extend the analysis beyond genre, decade, ratings and geography

Runtime vs Profitability & IMDb Rating



IMDb Votes vs Profit



QUERY 5 — RUNTIME

Movies in longer runtime tiers show higher average profit and higher average IMDb rating in this aggregation. The result describes an association, not a causal effect.

QUERY 6 — ENGAGEMENT

Among movies with more than 50k IMDb votes and positive profit, the regression line is upward. The scatter is wide, so audience attention alone does not explain profitability.

Data Quality & Validation

Quality checks validate consistency and reliability before the data is used for OLAP analysis.

01

Referential Integrity

Verify that fact records correctly reference dimensions and the movie genre bridge.

02

Profit Consistency

Check that profit is consistently derived from revenue and budget.

03

Rating Validation

Validate IMDb and TMDB rating values and identify missing or anomalous entries.

04

Reconciliation Coverage

Verify the cross source mapping performed through links.csv.

Quality checks ensure that the integrated data is consistent and reliable before OLAP analysis.

Conclusions & Future Work

The project delivers a reproducible analytical pipeline for integrating heterogeneous movie data and supporting multidimensional analysis.

What the project delivers

- Multi source integration**
Kaggle and IMDb data are integrated into a consistent analytical model.
- Reliable reconciliation**
links.csv provides an exact identifier mapping and reduces matching ambiguity.
- Multidimensional analysis**
The warehouse supports financial performance, rating comparison and temporal analysis.

Future Work

- 01 Extend the data scope**
Add streaming and box office data to enrich the analysis.
- 02 Expand analytical depth**
Investigate additional dimensions and performance indicators.
- 03 Improve reporting**
Develop dashboard oriented views for recurring OLAP analyses.

From heterogeneous raw data to a consistent warehouse for reproducible movie analytics.